

Notes: Brandt, Kambourov, and Storesletten (RES 2026)

Barriers to Entry and Regional Economic Growth in China

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1 Big picture

1.1 The empirical puzzle

In China’s manufacturing sector, *non-state* firms (NSOEs) display:

- very large cross-prefecture dispersion in wages and productivity in the mid-1990s, yet
- rapid *unconditional* convergence after 1995 in wages, output per worker, and measured TFP.

At the same time, locations with a larger state-owned enterprise (SOE) presence exhibit weaker non-state entry and weaker non-state entrant performance.

1.2 Core contribution

The paper builds a tractable, location-by-location Hopenhayn-style framework as an *accounting device*:

1. A static Hopenhayn model with heterogenous entrepreneurs and endogenous selection,
2. location-specific **capital wedges** and **output wedges** (as in misallocation work),
3. and a novel **entry barrier** modelled as a *license lottery* that blocks a fraction of potential entrepreneurs.

The model solves analytically and *aggregates*, so prefecture-level wage, TFP, and entry can be mapped to wedge objects.

1.3 Main findings

- The paper imputes prefecture-specific wedges and shows: **entry barriers explain most of the dispersion and convergence patterns**, especially in wages and TFP.
- Capital and output wedges move in the data, but they contribute little to explaining convergence in non-state performance.
- Entry barriers are strongly related to the **size of the SOE sector**; using IV strategies (including a Bartik-style instrument exploiting the 1997 SOE reform), the paper argues this relationship is **causal**.

The paper therefore links regional productivity growth to institutional constraints on private entry, mediated through the state sector.

2 Data, outcomes, and the empirical facts (Tables 1–2, Figures 1–3)

2.1 Data and units

Main data source: Chinese Industry Census (CIC) for 1995, 2004, and 2008 (manufacturing, firm-level output/value added, employment, wage bill, capital stock, etc.). NSOEs are defined by ownership categories (the paper discusses alternative definitions).

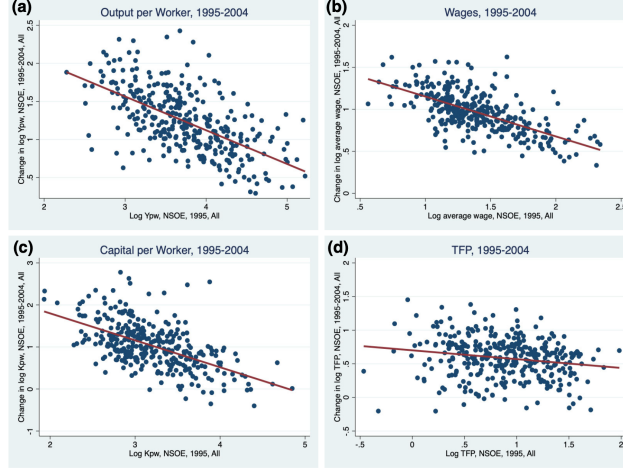


FIGURE 1
Convergence in the NSOE sector, 1995–2004

A location is a *prefecture* p (over 300 in the baseline sample). The empirical objects are constructed at the prefecture level for the non-state manufacturing sector.

2.2 Aggregate outcomes

Let $x_{p,t}$ denote a prefecture-level outcome:

$$x_{p,t} \in \left\{ \frac{Y_{p,t}}{N_{p,t}}, w_{p,t}, \frac{K_{p,t}}{N_{p,t}}, Z_{p,t} \right\},$$

where:

- Y/N is value added per worker (labour productivity),
- w is the average wage per worker,
- K/N is capital per worker,
- Z is measured aggregate TFP (Solow residual aggregated across industries within prefecture).

2.3 Unconditional β -convergence regression

The paper uses the standard convergence regression (Barro–Sala-i-Martin style). For horizon T :

$$\Delta \ln x_{p,t_0 \rightarrow t_0+T} \equiv \ln x_{p,t_0+T} - \ln x_{p,t_0} = - \left(1 - e^{-\beta T} \right) \ln x_{p,t_0} + \varepsilon_{p,t_0,t_0+T}. \quad (1)$$

The reported “annualized rate of β -convergence” is the implied β .

2.4 Interpretation: Table 1 and Figure 1

Figure 1 plots $\ln x_{p,1995}$ against $\Delta \ln x_{p,1995 \rightarrow 2004}$ for each outcome. The fitted line is negative in all panels, visually documenting unconditional convergence.

Table 1 quantifies:

TABLE 1
Dispersion and rates of unconditional convergence

	Labour productivity $\frac{Y}{N}$	Wage rate w	Capital per worker $\frac{K}{N}$	Aggr. TFP Z
Annualized rate of β -convergence 1995–2004	6.5%	7.2%	11.4%	2.6%
Annualized rate of β -convergence 1995–2008	6.4%	8.9%	23.6%	2.9%
90-10 ratio in 1995	4.4	2.3	3.6	3.1
90-10 ratio in 2004	3.0	1.8	3.1	3.9
90-10 ratio in 2008	3.2	2.1	3.9	4.3

TABLE 2
Comovements in wages, TFP, and firm entry

	1995			1995–2004			1995–2008		
	$\ln W$	$\ln TFP$	$\ln \Gamma$	$\Delta \ln W$	$\Delta \ln TFP$	$\Delta \ln \Gamma$	$\Delta \ln W$	$\Delta \ln TFP$	$\Delta \ln \Gamma$
$\ln W$	1.00			$\Delta \ln W$	1.00		1.00		
$\ln TFP$	0.42	1.00		$\Delta \ln TFP$	0.18	1.00	0.14	1.00	
$\ln \Gamma$	0.26	0.40	1.00	$\Delta \ln \Gamma$	0.24	0.09	1.00	0.25	1.00

- **Fast convergence in Y/N and w :** annualized rates around 6–9% depending on the period.
- **Very fast convergence in K/N :** especially over 1995–2008 (the table reports a very large annualized rate).
- **Slower convergence in TFP Z :** around 3% per year.
- **Dispersion patterns:** the 90–10 ratios indicate large dispersion in 1995; dispersion in some variables falls by 2004 but then rises by 2008 (dispersion can increase even with β -convergence when shocks matter).

2.5 Comovement of wages, TFP, and entry: Table 2

Define a prefecture entry rate $\Gamma_{p,t}$ as the employment share in *new* NSOE firms (firms founded within the last two years).

Table 2 reports correlations in levels and changes:

- In 1995, $\ln w$, $\ln Z$, and $\ln \Gamma$ are positively correlated.
- The same is broadly true in changes (1995–2004 and 1995–2008): places with faster wage growth also tend to have faster growth in entry and (to a lesser extent) measured TFP.

This positive comovement is a key empirical discipline for the model: the model must be able to move wages, TFP, and entry together.

2.6 State sector and non-state entrant performance: Figures 2–3

Let s_p be the size of the SOE sector in prefecture p (the paper uses SOE output share or SOE employment share).

Figure 2 shows that in 1995, NSOE *entrants* are weaker where the SOE sector is larger: lower wages, lower TFP, lower value added per worker, and lower capital per worker.

Figure 3 shows that NSOE entry rates are substantially lower in locations with a large SOE presence. These patterns motivate the later causal analysis linking SOEs $\rightarrow$ entry barriers $\rightarrow$ non-state performance.

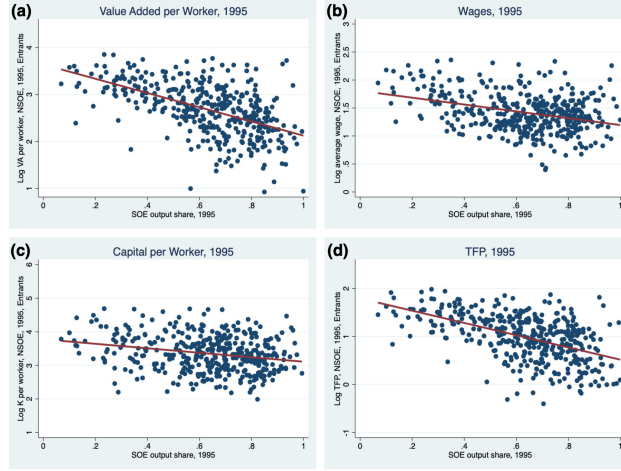


FIGURE 2
Characteristics of NSOE entrants in 1995

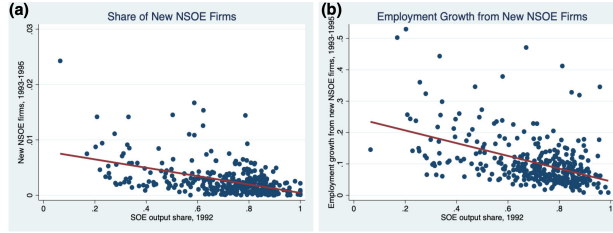


FIGURE 3
NSOE firm entry during 1993–5

2.7 A quick map from tables/figures to claims (revision checklist)

- **Figure 1 & Table 1:** document unconditional convergence (fast for w and Y/N , slower for Z) and show large initial dispersion.
- **Table 2:** shows positive comovement among wages, measured TFP, and entry across prefectures.
- **Figure 2:** shows NSOE entrant quality is lower where SOEs are larger (1995).
- **Figure 3:** shows NSOE entry is lower where SOEs are larger (early 1990s).
- **Table 3:** comparative statics in the benchmark model; entry barriers are the distortion that moves (w, Z, Γ) together.
- **Table 4 & Figure 4:** report and plot gross output/capital wedges and their relation to SOE presence.
- **Table 5:** documents worker and entrepreneur migration patterns used to measure N/M .
- **Figures 5–6:** show imputed entry barriers are higher in SOE-heavy prefectures; barriers fall over time.
- **Figure 7:** external validation using World Bank “Doing Business” measures.
- **Figures 8–9:** dynamic mechanism tests: wage growth is associated with younger firm populations and more new-firm employment shares.

- **Table 6:** convergence decomposition; entry barriers explain most of TFP (and much of wage) convergence.
- **Table 7:** aggregate growth decomposition; broad productivity growth dominates, but entry barriers contribute meaningfully.
- **Tables 8–10:** regressions linking SOEs to entry barriers and to growth/convergence (IV evidence).
- **Figure 10:** robustness of entry barrier estimates to within-prefecture wedge heterogeneity.

3 Benchmark model: a static Hopenhayn economy with wedges and an entry barrier

The paper’s benchmark model is static but has two stages: *entry* and *production*. The main purpose is to generate tractable expressions for wages, aggregate TFP, and entry as functions of wedges.

3.1 Environment and wedges (one location)

Consider a location (prefecture) p .

- Labour is location-specific and supplied inelastically: N_p workers are available to the NSOE sector.
- Capital is freely mobile across locations and rented at an exogenous rate $r + \delta$ (common across locations).
- There is a measure M_p of potential entrepreneurs.
- Each entrepreneur has a potential productivity draw z from a common distribution $f(z)$.
- A fixed operating cost $\nu > 0$ must be paid to operate a firm.

Distortions:

- **Output wedge** $\tau_{y,p}$: an implicit tax/subsidy on output revenues.
- **Capital wedge** $\tau_{k,p}$: an implicit tax/subsidy on capital costs.
- **Entry barrier** $\psi_p \in [0, 1)$: a fraction of potential entrepreneurs are denied a license. Only $(1 - \psi_p)$ get the option to operate.

In the benchmark, $\tau_{y,p}$ and $\tau_{k,p}$ are common across firms within p .

3.2 Technology

Each firm that operates chooses capital k and labour n and produces value added

$$y = z^{1-\eta} (k^{1-\alpha} n^\alpha)^\eta, \quad \eta \in (0, 1), \alpha \in (0, 1). \quad (2)$$

$\eta < 1$ delivers decreasing returns to scale in inputs; the technological labour income share is $\alpha\eta$.

3.3 Distribution of potential productivity

Potential productivity z is Pareto:

$$f(z) = \xi \underline{z}^\xi z^{-\xi-1}, \quad z \geq \underline{z}, \xi > 1, \quad (3)$$

where $\underline{z} > 0$ is the lower bound.

3.4 Firm problem (production stage)

Let the wage be w_p and define the effective rental cost

$$R_p \equiv (1 + \tau_{k,p})(r + \delta).$$

Conditional on having a license and choosing to operate, a firm solves:

$$\max_{k,n \geq 0} \Pi(z) = (1 - \tau_{y,p})y - w_p n - R_p k. \quad (4)$$

3.4.1 First-order conditions and factor shares

Using Cobb–Douglas structure, the FOCs imply the usual “revenue share” conditions:

$$w_p n^*(z) = \alpha \eta (1 - \tau_{y,p}) y^*(z), \quad (5)$$

$$R_p k^*(z) = (1 - \alpha) \eta (1 - \tau_{y,p}) y^*(z). \quad (6)$$

3.4.2 Closed form for optimal choices

A key property of (2) is constant returns in (z, k, n) , so optimal choices are linear in z :

$$y^*(z) = z \cdot \bar{y}_p, \quad (7)$$

$$n^*(z) = z \cdot \bar{n}_p = z \cdot \frac{\alpha \eta (1 - \tau_{y,p})}{w_p} \bar{y}_p, \quad (8)$$

$$k^*(z) = z \cdot \bar{k}_p = z \cdot \frac{(1 - \alpha) \eta (1 - \tau_{y,p})}{R_p} \bar{y}_p, \quad (9)$$

$$\Pi^*(z) = (1 - \tau_{y,p})(1 - \eta) z \cdot \bar{y}_p. \quad (10)$$

The scale term $\bar{y}_p$ depends only on prices and wedges:

$$\bar{y}_p = \left[\frac{(\eta(1 - \tau_{y,p}))^\eta \alpha^{\alpha\eta} (1 - \alpha)^{(1-\alpha)\eta}}{w_p^{\alpha\eta} R_p^{(1-\alpha)\eta}} \right]^{\frac{1}{1-\eta}}. \quad (11)$$

3.5 Entry decision and the productivity cutoff

Operating requires paying the fixed cost ν each period. Thus an entrepreneur operates iff $\Pi^*(z) \geq \nu$.

Define the cutoff z_p^* by $\Pi^*(z_p^*) = \nu$. Using (10),

$$z_p^* = \frac{\nu}{(1 - \tau_{y,p})(1 - \eta) \bar{y}_p}. \quad (12)$$

Substituting (11) into (12) yields a closed-form z_p^* in terms of $(w_p, R_p, \tau_{y,p})$ (this corresponds to equation (4) in the paper).

3.6 Labour market clearing

Only a fraction $(1 - \psi_p)$ of the M_p potential entrepreneurs draw a license and can enter. Among those, only productivity draws above z_p^* operate. Labour market clearing requires:

$$\int_{z_p^*}^{\infty} M_p(1 - \psi_p) n^*(z) f(z) dz = N_p. \quad (13)$$

Using $n^*(z) = z \bar{n}_p$ and Pareto (3),

$$\int_{z_p^*}^{\infty} z f(z) dz = \frac{\xi}{\xi - 1} \bar{z}^{\xi} (z_p^*)^{-(\xi-1)}.$$

Hence (13) becomes:

$$M_p(1 - \psi_p) \bar{n}_p \cdot \frac{\xi}{\xi - 1} \bar{z}^{\xi} (z_p^*)^{-(\xi-1)} = N_p. \quad (14)$$

Combining (14) with the expression for z_p^* and $\bar{n}_p$ delivers a log-linear wage equation.

3.7 Proposition 1 (equilibrium wage and aggregate TFP) and derivations

The paper states an analytical characterization; here is the same object with consistent notation.

Define:

$$\mu \equiv \frac{1}{1 - \eta + \xi \alpha \eta} > 0.$$

Proposition 1 (Equilibrium wage and Solow residual (benchmark model)). *In each location p , equilibrium wages and the measured Solow residual can be written as*

$$\begin{aligned} \ln w_p &= \mu(1 - \eta) \ln \left(\frac{(1 - \psi_p) \bar{z} M_p}{N_p} \right) + \mu \xi \ln(1 - \tau_{y,p}) \\ &\quad - \mu(1 - \alpha) \xi \eta \ln((1 + \tau_{k,p})(r + \delta)) + \Omega(\alpha_p, \eta, \xi, \nu), \end{aligned} \quad (15)$$

$$\begin{aligned} \ln Z_p &= \mu \alpha \eta (1 - \eta) \ln \left(\frac{(1 - \psi_p) \bar{z} M_p}{N_p} \right) - \mu(1 - \eta) \ln(1 - \tau_{y,p}) \\ &\quad + \mu(1 - \eta) [1 + (\xi - 1) \alpha \eta] \ln((1 + \tau_{k,p})(r + \delta)) + \hat{\Omega}(\alpha_p, \eta, \xi, \nu), \end{aligned} \quad (16)$$

where $\Omega(\cdot)$ and $\hat{\Omega}(\cdot)$ collect terms independent of the wedges. Moreover, w_p is decreasing in N_p/M_p , $\tau_{y,p}$, $\tau_{k,p}$, and ψ_p .

TABLE 3
Comparative statics of varying the wedges (τ_y, τ_k, ψ) and aggregate labour supply per potential entrepreneur, N/M

	$(1 - \tau_y)$	$(1 + \tau_k)$	$(1 - \psi)$	N/M
Wage rate w	+	−	+	−
Solow residual Z	−	+	+	−
Entry Γ	+	−	+	+

3.7.1 Derivation sketch (how to reproduce the wage equation)

The steps are:

1. Use (11) and (12) to express z_p^* as a power function of $(w_p, R_p, 1 - \tau_{y,p})$.
2. Use Pareto integrals to write total labour demand in (14) as a power function of z_p^* .
3. Substitute z_p^* into (14) and solve for w_p .
4. Take logs and collect coefficients. The elasticity terms in (15) are pinned down by the exponents in the Cobb–Douglas and the Pareto tail ξ .

The key intuition: the license barrier $(1 - \psi_p)$ scales down the mass of potential entrants, lowering labour demand and wages. Lower wages reduce the cutoff z_p^* , generating *negative selection* and lower measured TFP.

3.8 Entry rate and comparative statics (Table 3 and equation (8))

Define the measure of operating firms (entry mass, up to scaling):

$$\Gamma_p \equiv M_p(1 - \psi_p) \Pr(z \geq z_p^*).$$

Under Pareto, $\Pr(z \geq z_p^*) = (\underline{z}/z_p^*)^\xi$, so Γ_p is log-linear in wedges.

The paper's expression (equation (8)) can be written as:

$$\ln \Gamma_p = \mu(1-\eta) \ln((1 - \psi_p)\underline{z}M_p) + \mu\alpha\eta\xi \ln(N_p) + \mu\xi \ln(1 - \tau_{y,p}) - \mu\xi\eta(1-\alpha) \ln((1 + \tau_{k,p})(r + \delta)) + \tilde{\Omega}(\alpha_p, \eta, \xi, \nu). \quad (17)$$

3.8.1 How to read Table 3

Table 3 summarizes comparative statics (signs):

- Higher $(1 - \psi)$ (lower barrier) raises wages, measured TFP, and entry.
- Higher N/M lowers wages and TFP but raises entry (because lower wages relax the cutoff).
- A more favourable output wedge (higher $1 - \tau_y$) raises wages and entry but *reduces* measured TFP (it pushes up wages, tightening selection).
- A more adverse capital wedge (higher $1 + \tau_k$) lowers wages and entry but *raises* measured TFP (it reduces scale and wages; the induced selection changes the Solow residual mechanically).

A key empirical discipline is that in the data wages and TFP tend to comove positively. Table 3 implies that entry barriers are the clean distortion that naturally moves wages, TFP, and entry in the *same* direction.

4 Measuring wedges and entry barriers in the data (Tables 4–5, Figures 4–6)

The analytical aggregation results allow the paper to invert prefecture-level data into wedge objects.

4.1 Firm-level wedge formulas (equation (9))

From (5)–(6), the wedges satisfy:

$$1 - \tau_{y,p} = \frac{1}{\alpha\eta} \frac{w_p n_i}{y_i} \quad (\text{in the benchmark, common across } i \text{ in } p), \quad (18)$$

$$1 + \tau_{k,p} = \frac{1 - \alpha}{\alpha} \frac{w_p n_i}{(r + \delta)k_i}. \quad (19)$$

Empirically, the paper constructs *gross* wedge counterparts using firm-level data and then aggregates within prefectures accounting for industrial composition.

4.2 Prefecture-level gross output and capital wedges (equations (10)–(11))

Let j index industries. Let $Y_{j,p} = \sum_{i \in (j,p)} y_i$ and $Y_p = \sum_j Y_{j,p}$. Let $K_{j,p} = \sum_{i \in (j,p)} k_i$ and $K_p = \sum_j K_{j,p}$.

The paper's prefecture-level gross wedges can be summarized as weighted averages:

$$\Delta_p^y = \sum_j \left[\left(\sum_{i \in (j,p)} \frac{w_i n_i}{y_i} \cdot \frac{y_i}{Y_{j,p}} \right) \frac{1}{\alpha_j \eta} \right] \cdot \frac{Y_{j,p}}{Y_p}, \quad (20)$$

$$\Delta_p^k = \sum_j \left[\left(\sum_{i \in (j,p)} \frac{w_i n_i}{(r + \delta)k_i} \cdot \frac{k_i}{K_{j,p}} \right) \frac{1 - \alpha_j}{\alpha_j} \right] \cdot \frac{K_{j,p}}{K_p}. \quad (21)$$

(Exact weighting details match equations (10)–(11) in the paper; the main point is: aggregate within industry, then aggregate across industries using prefecture value-added/capital weights.)

TABLE 4
Moments for wedges

$\frac{WN}{Y}$	Output wedge r_y		$\frac{WN}{K}$	Capital wedge r_k		Entry barrier $\ln(1 - \psi)$		
(1)	avg. Δ_y (2)	std($\ln \Delta_y$) (3)	(4)	avg. Δ_k (5)	std($\ln \Delta_k$) (6)	avg. $\ln(1 - \psi)$ (7)	std($\ln(1 - \psi)$) (8)	
1995	0.28	1.00	0.32	0.15	1.00	0.36	-6.00	3.3
2004	0.22	0.79	0.26	0.17	1.16	0.28	-3.87	2.9
2008	0.21	0.76	0.37	0.20	1.33	0.45	-2.93	2.5

Notes: The table reports the mean and standard deviation of gross wedges and entry barriers across prefectures. The means for the gross output and capital wedges are normalized to unity in 1995. The entries in columns (1) and (4) refer to ratios calculated using aggregated data on the wage bill WN , capital K , and value added Y at the national level for non-state manufacturing firms. Data source: Chinese Industry Census.

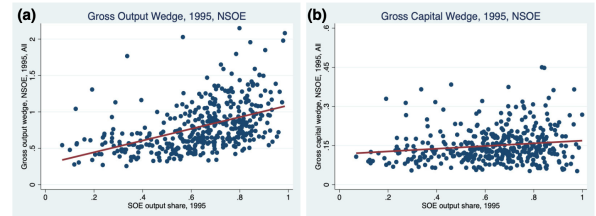


FIGURE 4
Gross output and gross capital wedges, 1995, all firms, NSOE sector
Notes: Each dot represents a prefecture. The left (right) panel plots the gross output (capital) wedge in the NSOE sector in 1995. The SOE output share in 1995 in each prefecture is on the horizontal axis. The top and bottom 1% observations in each panel are dropped.

TABLE 5
Migrant workers and entrepreneurs

	(1) $m_n \equiv \frac{\text{Migrant workers}}{\text{Total workers}}$	(2) $m_e \equiv \frac{\text{Migrant entrepreneurs}}{\text{Total entrepreneurs}}$	(3) $\text{corr}(m_n, m_e)$
1990	0.055	0.024	0.524
2000	0.249	0.115	0.484
2010	0.304	0.220	0.340

Notes: The table reports the share of workers and entrepreneurs who are migrants. Columns (1) and (2) show the aggregate shares of migrants. Column (3) reports the correlation between the worker migrant share and the entrepreneur migrant share across counties in China. Source: China Census. See [Online Appendix A](#) for details about measurements.

4.3 Interpreting Table 4 and Figure 4

Table 4 reports means and dispersions of these wedge objects over time:

- The average *gross output wedge* tends to fall over time (consistent with declining labour share).
- The average *gross capital wedge* rises over time.
- The average entry barrier declines over time (log gross entry barrier $\ln(1 - \psi)$ rises), and dispersion falls.

Figure 4 plots the 1995 gross wedges against the SOE output share: gross output wedges are strongly increasing in the SOE share (interpretable as subsidies/taxes patterns), while gross capital wedges are only weakly related. Importantly, the wedge patterns alone do not match the cross-sectional wage facts, motivating the focus on entry barriers.

4.4 Measuring N_p/M_p using migration data (equation (12))

To evaluate (15)–(17), one needs the effective labour supply per potential entrepreneur, N_p/M_p .

The paper uses Census evidence on migrants among workers and among entrepreneurs (proxied by CEOs).

Table 5 documents:

- migrants are prevalent (especially by 2010),
- migrant shares rise over time,
- worker and entrepreneur migrant shares are positively correlated across locations.

Under the paper’s measurement assumptions, the effective ratio can be written as:

$$\frac{N_p}{M_p} = \underbrace{(\text{NSOE share of manufacturing employment})}_{\text{removes SOE workers}} \times \underbrace{\frac{1 - m_{e,p}}{1 - m_{n,p}}}_{\text{locals among entrepreneurs / locals among workers}}, \quad (22)$$

where $m_{n,p}$ is the migrant share among workers and $m_{e,p}$ the migrant share among entrepreneurs.

4.5 Imputing entry barriers from wages (equation (13))

The key inversion step uses the wage equation to solve for entry barriers. Rearranging (15) yields:

$$\ln(1 - \psi_p) = \frac{1 - \eta + \xi\alpha\eta}{1 - \eta} \ln w_p - \frac{\xi}{1 - \eta} \ln \Delta_p^y + \frac{\xi\eta(1 - \alpha)}{1 - \eta} \ln \Delta_p^k + \ln\left(\frac{N_p}{M_p}\right) + \bar{\Omega}(\alpha_p, \eta, \xi, \underline{z}, \nu), \quad (23)$$

which is equation (13) in the paper (up to notational normalization).

4.6 Interpreting Figures 5–6

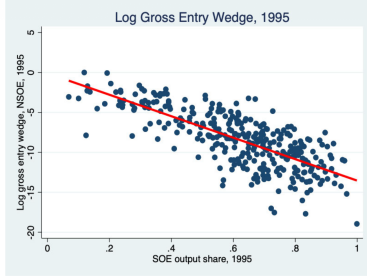


FIGURE 5

Log gross entry barriers, $\ln(1 - \psi)$, 1995

Notes: Each dot represents a prefecture. The figure plots the log gross entry barrier in the NSOE sector in 1995, and the solid line is the fitted regression line. The 1995 SOE output share in a prefecture is on the horizontal axis.

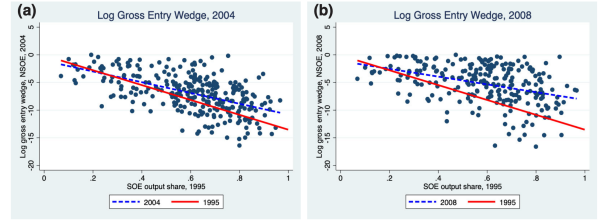


FIGURE 6

Log gross entry barriers, $\ln(1 - \psi)$, 2004 and 2008

Notes: Each dot represents a prefecture. The left (right) panel plots the log gross entry barrier in the NSOE sector in 2004 (2008), and the dotted line is the corresponding fitted regression line. The solid line is the fitted regression line for the log gross entry barrier in 1995. The 1995 SOE output share in a prefecture is on the horizontal axis.

Figures 5–6 plot the imputed entry barrier against the 1995 SOE share:

- In 1995 there is a strong negative relationship between $\ln(1 - \psi)$ and SOE share: larger SOE presence $\Rightarrow$ tighter entry barrier.
- The relationship persists but weakens by 2004 and 2008, consistent with SOE reforms reducing barriers especially in places with initially large SOE sectors.

5 Validation and mechanism evidence (Figures 7–9 and selection facts)

The benchmark model has an identification caveat: in a static Pareto setting, proportional shifts in the productivity distribution and changes in $(1 - \psi)$ can look similar in aggregate moments. The paper addresses this with external and dynamic evidence.

5.1 Direct validation: World Bank “Doing Business” (Figure 7)

Figure 7 shows strong correlations between the paper’s entry barrier measure and the World Bank’s provincial measures of:

- rank for ease of starting a business,
- days to start a business,
- cost of starting a business (percent of GDP per capita).

Interpretation: prefectures/provinces measured to have tighter entry barriers by the model are also those where direct institutional measures suggest entry is more costly.

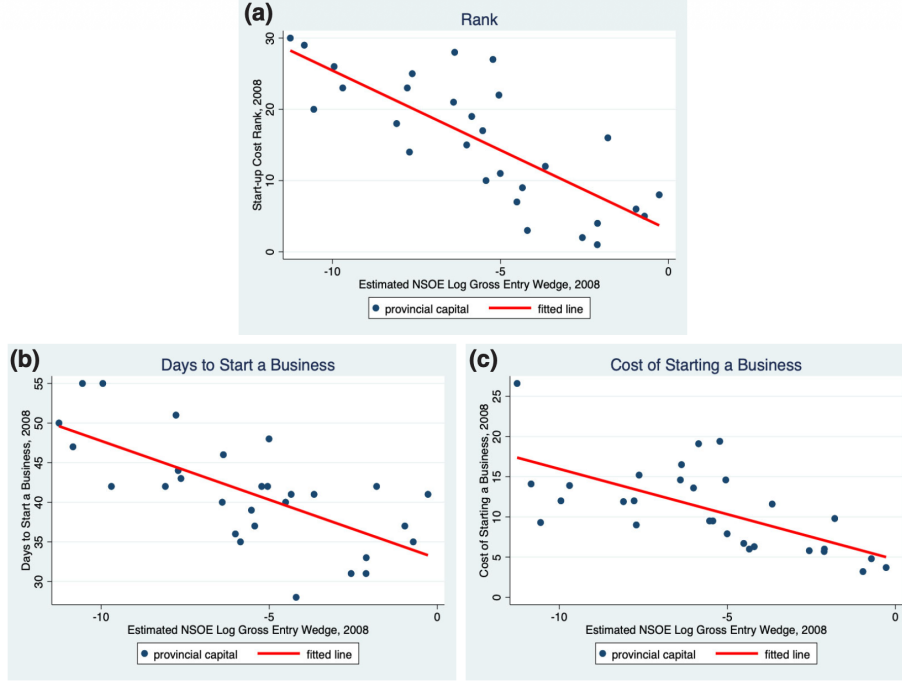


FIGURE 7
“Doing business in China” and entry barriers, 2008

Notes: Each dot represents a provincial capital city or a centrally administered municipality. Each panel shows a scatter plot of the estimated log gross entry barrier $\ln(1 - \psi)$ against a World Bank measure of the cost of doing business in China in 2008: rank (top panel), days to start a business (bottom left panel), and cost of starting a business (bottom right panel). The solid line is the fitted regression line.

5.2 Dynamic extension and a separating implication (equations (14)–(15))

To distinguish entry barriers from general TFP shifters, the paper introduces a two-period extension:

- Licenses are permanent; all incumbents survive to period 2.
- The period-2 entry barrier ψ_2 applies *only* to new potential entrants.
- Other shocks (TFP shifter Z_t , wedges, labour supply) affect incumbents and entrants symmetrically in period 2.

This yields a sharp prediction: **average firm age depends on the entry barrier only.**

Let A_t denote the (employment-weighted) average firm age. In the two-period setup, all period-1 firms have age $A_1 = 1$ and survive into period 2. The period-2 firm population is therefore a mixture of incumbents (age 2) and new entrants (age 1), implying

$$\frac{A_2}{A_1} = 1 + \frac{1 - \psi_1}{(1 - \psi_1) + (1 - \psi_2)}. \quad (24)$$

This is equation (14) in the paper: holding other shocks fixed, a higher period-2 barrier ψ_2 mechanically increases average age because it reduces the mass of new firms.

The same extension yields an explicit wage-growth decomposition (equation (15) in the paper). To avoid notational confusion with the Solow residual Z_p , denote the *common* TFP shifter by Z_t .

Then

$$\ln w_2 - \ln w_1 = \mu(1-\eta) \left[\xi \ln \left(\frac{Z_2}{Z_1} \right) - \ln \left(\frac{N_2}{N_1} \right) + \ln \left(1 + \frac{1-\psi_2}{1-\psi_1} \right) \right] + \mu \xi \ln \left(\frac{1-\tau_{y,2}}{1-\tau_{y,1}} \right) - \mu(1-\alpha) \xi \eta \ln \left(\frac{1+\tau_{k,2}}{1+\tau_{k,1}} \right). \quad (25)$$

Thus wages alone cannot identify entry barriers, but *wage growth jointly with age/entry composition can*.

5.3 Empirical mechanism tests: Figures 8–9

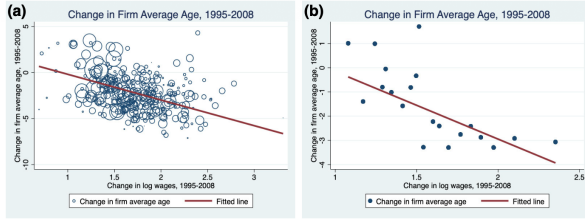


FIGURE 8
Change in firm average age, by prefecture, 1995–2008

Notes: The figures plot the change in firm average age in prefecture between 1995 and 2008 as a function of the 1995–2008 change in log wages in a prefecture. The left panel plots the figure for every prefecture in the sample. The right panel groups prefectures into 20 bins, sorted on wage growth. Each bin represents 5% of aggregate employment.



FIGURE 9
Employment share in new firms, by prefecture, 1995–2008

Notes: The figures plot the change in the employment share of new firms in a prefecture between 1995 and 2008 as a function of the 1995–2008 change in log wages in a prefecture. The left panel plots the figure for every prefecture in the sample. The right panel groups prefectures into twenty bins sorted on wage growth. Each bin represents 5% of aggregate employment.

Figure 8 shows a negative relationship between wage growth and changes in average firm age: locations with higher wage growth end up with *younger* firms, consistent with falling entry barriers.

Figure 9 shows a positive relationship between wage growth and the increase in the employment share of new firms: places with higher wage growth exhibit more activity in young cohorts.

These patterns support the interpretation that the main measured distortion is indeed an entry barrier operating through firm entry and selection.

5.4 TFP distribution evidence for selection

The benchmark assumption is that the distribution of *potential* entrepreneurial productivity is common across locations. Then differences in outcomes are driven by different selection cutoffs z_p^* .

Implications:

- High-wage locations should have a thinner *lower tail* of the observed firm TFP distribution (because z_p^* is higher).
- Upper tails should be similar across high- and low-wage locations (selection truncates the bottom, not the very top).

The paper reports this evidence in online appendix figures: lower tails differ as predicted, and upper tails look similar and approximately Pareto.

6 Accounting for convergence and growth (Tables 6–7)

Now the paper uses the benchmark model as an accounting system: given measured wedges over time, how much convergence and growth can be attributed to each component?

6.1 Method: counterfactual convergence using the model

Use (15) and (16) to compute counterfactual $w_{p,t}$ and $Z_{p,t}$. To attribute convergence, hold all factors at 1995 levels except one factor (e.g. entry barrier) and recompute the implied 2004 (or 2008) outcomes. Then run the same β -convergence regression (1) on the counterfactual outcomes.

6.2 Interpreting Table 6 (convergence decomposition)

Table 6 reports annual convergence rates in TFP and wages for:

- the full model (“All” row), and
- counterfactual scenarios where only one factor changes (entry barrier, wedges, labour share, N/M), plus a decomposition of N/M into migration and NSOE-share components.

The headline result:

- **Entry barriers account for essentially all convergence in TFP.**
- Entry barriers also explain a large share of convergence in wages.
- Output wedges can contribute to TFP convergence but tend to push wages in the opposite direction (as predicted by the benchmark comparative statics).
- Changes in N/M have small net effects because worker migration, entrepreneur migration, and NSOE-share movements offset each other.

TABLE 6
Annual rate of convergence in TFP and wages: 1995–2004 and 1995–2008

Change in	TFP		Wages	
	1995–2004	1995–2008	1995–2004	1995–2008
All	0.028	0.023	0.068	0.075
$(1 - \psi)$	0.029	0.027	0.019	0.032
$(1 + \tau_k)$	−0.005	−0.004	0.015	0.019
$(1 - \tau_y)$	0.008	0.008	−0.001	−0.016
$a\eta$	−0.004	−0.003	0.020	0.019
N/M	−0.001	−0.001	0.003	−0.001
—	0.002	0.001	0.009	0.008
—	−0.000	−0.000	−0.003	−0.004
— NSOE share	−0.002	−0.002	−0.005	−0.005

Notes: The table reports the annual rate of convergence in TFP and wages across prefectures for the 1995–2004 and 1995–2008 time periods. The β -convergence coefficient for prefectures p between times t_0 and $t_0 + T$ is estimated from the regression $(\frac{1}{T}) \ln(\frac{Z_{p,t_0+T}}{Z_{p,t_0}}) = a - (\frac{1 - e^{-\beta T}}{T}) \ln(y_{p,t_0}) + u_{p,t_0,t_0+T}$, where u_{p,t_0,t_0+T} represents an average of error terms, $u_{p,t}$, between times t_0 and $t_0 + T$. Each row in the table reports what the convergence in TFP and wages would have been had only one of the listed variables changed. The row “all” allows all factors to change and captures by construction the estimated empirical convergence rate. The top and bottom 1% observations for each variable change are dropped.

TABLE 7
Increase in Y , Y/N , TFP, and wages: 1995–2008

Change in log	Y	Y/N	TFP	w
	2.35	1.89	1.13	1.47
Fraction (in per cent) accounted for by:				
$(1 - \psi)$	19	24	20	31
z	74	91	75	116
$(1 - \tau_y)$	−6	−7	5	−27
$(1 + \tau_k)$	−9	−11	6	−12
N/M	22	2	−7	−10

Notes: The table reports the share of aggregate growth that can be accounted for by changes in each of the factors.

6.3 Interpreting Table 7 (aggregate growth decomposition)

Table 7 decomposes aggregate growth (1995–2008) in Y , Y/N , TFP, and wages into contributions from:

- the entry barrier $(1 - \psi)$,
- a common productivity shifter (labelled z in the table, capturing broad TFP growth),
- wedges $(1 - \tau_y)$ and $(1 + \tau_k)$,
- and N/M .

The main takeaway:

- broad productivity growth accounts for the bulk of aggregate growth (as expected),
- but entry barriers are the second most important driver, with economically meaningful contributions to TFP and wage growth.

7 The SOE sector and entry barriers: regressions and identification (Tables 8–10)

Since entry barriers dominate the accounting, the paper then asks: *what drives entry barriers?* The central hypothesis is that the SOE sector raises barriers to private entry.

7.1 Cross-sectional specification (equation (16))

For each year t , estimate:

$$\ln(1 - \psi)_{p,t} = \beta_0 + \beta S_{p,t} + X'_{p,t}\gamma + \varepsilon_{p,t}, \quad (26)$$

where $S_{p,t}$ is SOE employment share, and $X_{p,t}$ includes controls (education, agriculture share, labour force participation; and fiscal revenue per government worker $\ln FREV$ in some years).

7.2 Interpreting Table 8 (cross-section OLS and IV)

Table 8 reports OLS and several IV designs:

- IV using lagged SOE share,
- IV using 1978-era SOE share constructed from firms established before 1978,
- IV using province-level 1978 SOE output shares.

Reading the coefficients:

- The SOE share has a **large negative** association with $\ln(1 - \psi)$ in all years: more SOEs $\Rightarrow$ smaller $(1 - \psi) \Rightarrow$ tighter entry barriers.
- IV estimates are typically larger in magnitude than OLS, consistent with attenuation from measurement error in SOE share (especially in 2004 when ownership coding becomes noisier).
- Fiscal capacity ($\ln FREV$) is positively related to $\ln(1 - \psi)$ in some specifications: richer local governments are associated with lower entry barriers.

7.3 Panel/difference specification and Bartik instrument (equation (17))

To study changes:

$$\Delta \ln(1 - \psi)_{p,t} = \beta_0 + \beta \Delta S_{p,t} + \Delta X'_{p,t}\gamma + \Delta \varepsilon_{p,t}. \quad (27)$$

Endogeneity concerns remain (unobserved shocks; reverse causality; measurement error). The paper therefore uses a Bartik-style instrument exploiting the 1997 SOE reform: predicted local SOE employment changes are constructed from national industry-level SOE employment changes weighted by the local 1995 SOE industry composition and scaled by initial SOE importance.

TABLE 8
The entry barrier in 1995, 2004, and 2008

	$\ln(1 - \psi)$	OLS	IV _{lag}	IV ₁₉₇₈	IV _{prov}
1995	S	-12.3*** (1.16)	-15.4*** (1.44)	-13.3*** (1.46)	-13.4*** (3.97)
	$\ln \text{FREV}$	1.44*** (0.40)	0.97** (0.42)	1.29*** (0.42)	0.91 (1.28)
	Controls	Yes	Yes	Yes	Yes
First stage:	IV coefficient		0.72*** (0.03)	0.94*** (0.05)	1.07*** (0.18)
	st. error				
	R^2		0.75	0.73	0.77
2004	S	-9.9*** (1.30)	-14.7*** (2.09)	-16.7*** (2.37)	-19.6*** (5.77)
	$\ln \text{FREV}$	1.61*** (0.48)	1.08** (0.52)	0.86 (0.54)	0.88 (0.85)
	Controls	Yes	Yes	Yes	Yes
First stage:	IV coefficient		0.62*** (0.05)	0.70*** (0.07)	0.85*** (0.22)
	st. error				
	R^2		0.54	0.48	0.64
2008	S	-11.1*** (1.35)	-11.5*** (1.56)	-15.2*** (2.33)	-8.3* (4.66)
	Controls	Yes	Yes	Yes	Yes
First stage:	IV coefficient		0.88*** (0.03)	0.70*** (0.06)	1.04*** (0.26)
	st. error				
	R^2		0.78	0.44	0.50

Notes: The table reports the OLS and IV results from a regression of the log gross entry barrier on the SOE employment share (S) and fiscal revenues per government worker (FREV) in a prefecture in 1995. Controls include average educational attainment, agricultural employment share, and labour force participation rate in 1990. S is available in all years. FREV is available in 1995 and 2004. Standard errors are in parentheses. ***Statistically significant at 1%; **statistically significant at 5%; *statistically significant at 10%.

TABLE 9
Change in the entry barrier and the SOE sector, 1995–2004

$\Delta \ln(1 - \psi)$	OLS	OLS	IV _{Bartik}	IV _{Bartik}
ΔS	-1.58 (1.16)	-1.44 (1.28)	-7.41*** (2.61)	-8.31*** (2.87)
$\Delta \ln \text{FREV}$		1.30*** (0.40)	0.91** (0.44)	0.91** (0.44)
First stage				
IV coefficient			0.61*** (0.07)	0.64*** (0.08)
st. error				
R^2			0.21	0.24

Notes: The table reports the OLS and IV results from a regression of the change in the log gross entry barrier on the change in the SOE employment share (ΔS) and in the log fiscal revenues per government worker ($\Delta \ln \text{FREV}$) in a prefecture between 1995 and 2004. Standard errors are in parentheses. ***Statistically significant at 1%; **statistically significant at 5%.

7.4 Interpreting Table 9 (Bartik IV)

Table 9 shows:

- OLS in differences yields a negative relationship: where SOE employment share falls more, entry barriers fall more.
- Bartik IV yields a much larger magnitude, interpreted as causal: SOE downsizing reduces entry barriers.

7.5 Interpreting Table 10 (SOEs, convergence, and growth)

Table 10 regresses 1995–2004 growth in NSOE outcomes (wages, value added per worker, TFP, capital per worker) on:

- instrumented SOE-sector growth (or decline) and
- initial 1995 levels (to capture conditional convergence), with province fixed effects.

The key interpretation:

- conditional convergence remains strong (coefficients on initial levels are negative),
- and instrumented SOE employment decline has economically significant effects on NSOE wage growth, labour productivity growth, and capital deepening.

This supports the narrative: SOE reform → reduced entry barriers → improved private-sector performance and convergence.

8 Political economy wedge determination with SOEs (Proposition 2 and equation (18))

To rationalize why SOEs map into entry barriers, the paper provides an extension in which local officials (cadres) choose wedges subject to an SOE employment target.

TABLE 10
NSOE growth and the role of the state sector, 1995–2004

	$\Delta \ln w$	$\Delta \ln(VA/N)$	$\Delta \ln TFP$	$\Delta \ln(K/N)$
$\Delta \hat{S}_p$	−0.52*** (0.18)	−1.21*** (0.29)	−0.36 (0.26)	−1.87*** (0.39)
$\ln x_{1995}$	−0.51*** (0.04)	−0.44*** (0.05)	−0.36*** (0.06)	−0.51*** (0.07)
Province F.E.	Yes	Yes	Yes	Yes
First stage				
β_{IV}	0.83***	0.87***	0.87***	0.88***
s.e.	(0.09)	(0.09)	(0.09)	(0.09)
adj. R^2	0.38	0.36	0.36	0.36

Notes: The table reports 1995–2004 growth in average wages, value added (VA) per worker, TFP, and log capital per worker for NSOE firms across prefectures. The role of the state sector growth is instrumented by IV_{Bartik} . Notes: Standard errors are in parentheses. ***Statistically significant at 1%.

8.1 Key constraint linking wedges to SOE employment

Let $\bar{\Lambda}_{SOE}$ denote an exogenous target for the SOE employment share. Market clearing plus the model implies a relationship:

$$\frac{1 - \bar{\Lambda}_{SOE}}{\bar{\Lambda}_{SOE}} \cdot \frac{1}{1 - \psi} = (1 - \tau_y)^{\frac{\xi}{1-\eta}} \left(\frac{1}{1 + \tau_k} \right)^{\frac{\xi(1-\alpha)\eta}{1-\eta}}. \quad (28)$$

which is the content of equation (18) in the paper: increasing any wedge reduces NSOE labour demand and raises equilibrium SOE employment, helping satisfy the target.

8.2 Proposition 2 (why the cadre uses entry barriers)

Given non-negativity of wedges and an objective aligned with maximizing profits of a representative NSOE entrepreneur (crony capitalism motivation), the paper shows:

Proposition 2 (Optimal wedge choice under an SOE employment constraint). *The constrained optimal choice of wedges sets the output and capital wedges to zero, $\tau_y = \tau_k = 0$, and uses a strictly positive entry barrier $\psi > 0$ to satisfy the SOE employment target. Moreover, an exogenous increase in the SOE employment target implies a higher entry barrier.*

Intuition. Output and capital wedges distort firm scale and reduce profits directly. An entry barrier reduces the number of competitors and can satisfy the SOE employment requirement while preserving profits for the favoured entrant. Thus the political economy mechanism naturally generates a strong link between SOE presence and entry barriers.

9 Extension: heterogeneous firm-level wedges (Propositions 3–4, Figure 10)

The benchmark assumes common wedges within a prefecture. The paper extends the model to allow *firm-specific* output and capital wedges within each prefecture and shows the main conclusion—entry barriers dominate the accounting—is robust.

9.1 Potential wedge distribution (equation (19))

Each potential entrepreneur draws (z, τ_k, τ_y) before deciding to operate. Potential productivity z is still Pareto and independent of wedges, but selection induces correlation between wedges and observed productivity among operating firms.

To keep aggregation tractable, potential wedges are assumed jointly log-normal within a prefecture. Let $(\bar{\tau}_k, \bar{\tau}_y)$ denote the (prefecture-specific) *mean* wedges and let $(\sigma_k, \sigma_y, \sigma_{ky})$ denote the variance-covariance terms of log wedges:

$$\begin{aligned}\mathbb{E}[\ln(1 + \tau_k)] &= \ln(1 + \bar{\tau}_k) - \frac{\sigma_k}{2}, \\ \mathbb{E}[\ln(1 - \tau_y)] &= \ln(1 - \bar{\tau}_y) - \frac{\sigma_y}{2}, \\ \text{Var}[\ln(1 + \tau_k)] &= \sigma_k, \quad \text{Var}[\ln(1 - \tau_y)] = \sigma_y, \quad \text{Cov}[\ln(1 + \tau_k), \ln(1 - \tau_y)] = \sigma_{ky}.\end{aligned}\quad (29)$$

These are mean-preserving spreads: $\mathbb{E}[1 + \tau_k] = 1 + \bar{\tau}_k$ and $\mathbb{E}[1 - \tau_y] = 1 - \bar{\tau}_y$. Setting $\sigma_k = \sigma_y = \sigma_{ky} = 0$ nests the benchmark model.

9.2 Proposition 3: equilibrium wage and Solow residual with wedge dispersion

Let $\mu \equiv 1/(1 - \eta + \xi\alpha\eta)$ as before. Proposition 3 in the paper shows:

$$\begin{aligned}\ln w &= \mu(1 - \eta) \ln \left[(1 - \psi) \bar{z} \frac{M}{N} \right] + \mu\xi \ln(1 - \bar{\tau}_y) - \mu\xi\eta(1 - \alpha) \ln \left((1 + \bar{\tau}_k)(r + \delta) \right) + \Omega \\ &\quad + \mu\xi \left(\frac{\xi}{1 - \eta} - 1 \right) \frac{\sigma_y}{2} + \mu\xi\eta(1 - \alpha) \left(\frac{\xi\eta(1 - \alpha)}{1 - \eta} + 1 \right) \frac{\sigma_k}{2} - \mu\xi^2(1 - \alpha) \frac{\eta}{1 - \eta} \sigma_{ky},\end{aligned}\quad (30)$$

$$\begin{aligned}\ln Z &= \mu\alpha\eta(1 - \eta) \ln \left[(1 - \psi) \bar{z} \frac{M}{N} \right] - \mu(1 - \eta) \ln(1 - \bar{\tau}_y) + \mu(1 - \eta)(1 - \alpha\eta(1 - \xi)) \ln(1 + \bar{\tau}_k) + \hat{\Omega} \\ &\quad - (\xi - (1 - \eta)) \left(\frac{1}{1 - \eta} + \mu \right) \frac{\sigma_y}{2} - \left[2(1 - \eta)(1 - \alpha\eta) + \alpha\xi\eta(2 - \eta(1 + \alpha)) \right] \frac{\mu(1 - \eta + (1 - \alpha)\xi\eta)}{(1 - \eta)} \frac{\sigma_k}{2} \\ &\quad + \left((1 - \eta)(\eta(1 - \alpha) + (\xi - 1)\alpha\eta + 1) + \alpha\eta^2(1 - \alpha)\xi \right) \frac{\mu\xi}{1 - \eta} \sigma_{ky}.\end{aligned}\quad (31)$$

The paper highlights the qualitative comparative statics:

- w is decreasing in ψ , N/M , $\bar{\tau}_k$, and $\bar{\tau}_y$, but increasing in σ_y , σ_k , and $-\sigma_{ky}$.
- Z is decreasing in ψ , N/M , σ_y , σ_k , and $-\sigma_{ky}$, but increasing in $\bar{\tau}_k$ and $\bar{\tau}_y$.

Intuition: dispersion in wedges reallocates size toward subsidized firms, raising labour demand (higher wages) but generating negative selection that lowers the Solow residual.

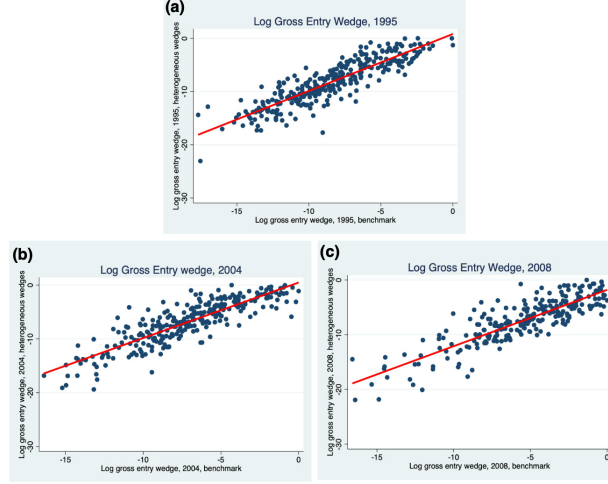


FIGURE 10

Log gross entry barriers, benchmark model, and model with wedge heterogeneity

Notes: Each dot represents a prefecture. The graphs plot the log gross entry barriers in the benchmark model and in the model with heterogeneous wedges in 1995, 2004, and 2008. The solid line is the fitted regression line.

9.3 Proposition 4: identifying wedge dispersion from observed moments

Even though entry is selective, the joint log-normal wedge parameters can be identified from first and second moments of observed wedges among operating firms ($z \geq z^*$). Proposition 4 provides:

$$\frac{\text{std}\{(1 + \tau_k)(r + \delta) \mid z \geq z^*\}}{\mathbb{E}\{(1 + \tau_k)(r + \delta) \mid z \geq z^*\}} = \sqrt{\exp(\sigma_k) - 1}, \quad (32)$$

$$\frac{\text{std}\{(1 - \tau_y) \mid z \geq z^*\}}{\mathbb{E}\{(1 - \tau_y) \mid z \geq z^*\}} = \sqrt{\exp(\sigma_y) - 1}, \quad (33)$$

$$\frac{\text{cov}\{(1 + \tau_k)(r + \delta), (1 - \tau_y) \mid z \geq z^*\}}{\mathbb{E}\{(1 - \tau_y) \mid z \geq z^*\} \cdot \mathbb{E}\{(1 + \tau_k)(r + \delta) \mid z \geq z^*\}} = \exp(-\sigma_{ky}) - 1, \quad (34)$$

$$\mathbb{E}\{(1 - \tau_y) \mid z \geq z^*\} = \exp\left(\ln(1 - \bar{\tau}_y) + 2\frac{\xi}{1 - \eta}\frac{\sigma_y}{2} - \left(\xi(1 - \alpha)\frac{\eta}{1 - \eta}\right)\sigma_{ky}\right), \quad (35)$$

$$\mathbb{E}\{(1 + \tau_k)(r + \delta) \mid z \geq z^*\} = \exp\left(\ln[(1 + \bar{\tau}_k)(r + \delta)] + 2\xi(1 - \alpha)\frac{\eta}{1 - \eta}\frac{\sigma_k}{2} - \frac{\xi}{1 - \eta}\sigma_{ky}\right). \quad (36)$$

Given these identified wedge moments and the observed wage w , one can again invert (30) to back out the entry barrier ψ .

9.4 Interpretation of Figure 10

Figure 10 plots (for 1995, 2004, and 2008) the log gross entry barriers from the benchmark model against those from the heterogeneous-wedge model. The correlations are high (around 0.86–0.89), implying that the benchmark entry-barrier estimates are robust to allowing within-prefecture wedge dispersion.

10 Summary

- Empirically, non-state manufacturing shows fast prefecture-level convergence in wages and productivity after 1995.
- A Hopenhayn model with selection can connect entry rates, wages, and aggregate TFP in a coherent way.
- Among wedges, an **entry barrier** is the distortion that naturally moves (w, Z, Γ) together.
- Quantitatively, entry barriers account for most of the convergence; capital/output wedges play a smaller role.
- The state sector is a key upstream determinant: SOE presence raises entry barriers; SOE reform reduced them and accelerated private-sector performance.